

Smart Waste Collection System Personas and Pain Points

Bellevue Smart Waste Management System Persona and Pain Point Analysis

Persona 1: Garbage Collection Officer



Name : Paul (35 years old)

Occupation : District Operations Manager

Housing type : Small apartment near Bellevue

Income level : Middle class

Responsibilities

- Regularly collect household waste bins within designated areas
- Clean up overflowing waste and surrounding areas during collection
- Visit an average of 80 to 120 bins per day

! Pain Points

- **Lack of prior information:** All bins visited under the same criteria (inefficient), approximately 35% of daily collection points were in a state where collection was unnecessary.
- **Workload Increase:** Stress escalates due to exceeding planned time and hygiene issues caused by cleaning up overflowing trash.
- **Physical strain:** Energy expenditure due to unnecessary movement

✓ Needs

- I want to selectively visit only the places where collection is truly necessary.
- Collect before overflowing to minimize additional cleanup work
- Minimize unnecessary movement to efficiently manage daily work routes

Persona 2: City Government Environmental Policy Manager



Name : Sara (50 years old)

Occupation : Environmental Policy Manager

Housing type : Single-family home in Bellevue

Income level : Upper class

Responsibilities

- Cleanliness policy implementation and public complaint handling
- Cost/performance management
- Ensuring residents experience a clean urban environment
- Minimizing overflow incidents

! Pain Points

- **Surge in complaints:** Garbage overflow complaints on the rise (Cleanliness satisfaction plummets from 62% to 48%)
- **Damage to City Image:** Concerns over declining tourist numbers due to a tarnished city image, damage to the city's reputation caused by the spread of the #Dirty hashtag, and a target for political attacks.
- **Pressure to demonstrate results:** Need to prove visible outcomes relative to budget, bear failure risk when adopting new technologies, face political pressure
- **Sanitation Issues:** Expanding sanitation and public health problems due to the emergence of harmful animals (raccoons, rats)

✓ Needs

- Demonstrate policy outcomes through data-driven reporting and promote achievements in realizing “smart cities”
- Reducing Crisis Situations Through Overflow Prediction
- Cost reduction + Increased public trust

Persona 3: Ordinary Citizens



Name : Charlie (23 years old)

Occupation : college student

Housing type : Single-family home in Bellevue

Income level : Middle class

Daily Routine

- Walking commute to campus (20 minutes)
- Park running/walking 3-4 times per week
- Weekend trekking and outdoor activities

! Pain Points

- **Unpredictability:** The frustration and wasted effort caused by not knowing the location of trash cans or their “availability status.”
- **Hygiene concerns:** Blocked entrances, litter left around, summertime odors and insects, discomfort caused by foreign matter on surfaces
- **Value gap:** Despite strong environmental awareness, the lack of infrastructure ultimately forces people to carry trash in bags, causing inconvenience.
- **Insufficient information:** During events/festivals when trash volume surges, the collection system fails to operate, leaving waste uncollected.

✓ Needs

- Clean and hygienic trash can management
- A digital map/app that allows real-time checking of location and current occupancy levels

Persona 4: Collection Operations Manager



Name : Alex (45 years old)

Occupation : District Operations Manager

Housing type : Seattle area apartment

Income level : Upper-middle class

Responsibilities

Collection Schedule & Route Design: Daily/weekly schedule management for 15 routes, optimized using Google Maps + GPS data

- Vehicle Driver Assignment: 5 AM driver briefing, real-time adjustments based on weather and other conditions

- Complaint Handling: Processing 50-100 cases of "missed garbage collection/overflow" received at customer center

- Performance Monitoring & Budget Management: Collection tonnage, fuel consumption, monthly \$1.2M operating costs (labor 65%, fuel 20%, vehicle maintenance 10%)

! Pain Points

- **Workload paralysis:** Surge in complaints after holidays/heavy rain (over 100 calls daily), repeated issues due to insufficient time for route redesign -> Same complaints repeated the next day
- **Inefficient Prioritization:** Lack of real-time load data, prioritizing only based on complaint locations, duplicate (25%)/unnecessary (35%) collections
- **Management Dilemma:**
 - Reducing collection frequency saves costs -> Leads to a surge in complaints -> Risk of contract cancellation
 - Daily fixed collection -> Exceeds fuel/labor cost budget -> Criticism from headquarters for budget management failure

✓ Needs

- Real-time Saturation Monitoring System and Data-Driven Routing Optimization Solution
- Predictive system enabling proactive response before civil complaints arise
- A solution that achieves both cost savings and service quality

Pain Points Summary

Persona	Pain Points	Needs	Value of the Smart System
Paul (Field Worker)	Unable to identify collection needs in advance (inefficiency), additional cleanup due to overflowing bins, physical strain and injury risk	Visit only necessary bins with optimized routes	Reduced workload and improved efficiency
Sara (City Official)	Surge in complaints due to overflowing bins, damage to city image, administrative trust decline, pressure to prove performance	Data-based evidence and overflow prevention	Transparent performance and public trust
Charlie (Citizen)	Lack of information on bin capacity, discomfort due to odor and unsanitary conditions, difficulty practicing eco-friendly behavior	Clean bins and real-time status info	Better convenience and quality of life
Alex (Operations Manager)	Workflow disruption due to excessive complaints, inefficient prioritization without real-time data, trade-off between cost and service quality	Real-time monitoring and route optimization	Lower labor costs and fewer complaints

Hypotheses to Validate

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Hypothesis 1

Early customers respond more strongly to cost reduction, complaint reduction, and risk mitigation than to environmental values.

Hypothesis 2

If trash bin fill-level and status information can be checked in advance, waste collection workers will reduce unnecessary travel and physical effort, and therefore are likely to respond positively to this solution.

Hypothesis 3

By monitoring trash bin fill levels in real time and managing waste volume as data, collection crews can visit only locations that truly require service and follow more efficient routes. This will enable better daily route and time management, reduce unnecessary and duplicate collections, and ultimately lower labor costs.

MVP Development Plan

“Minimum feature implementation to validate information
value and service demand”

System Components & Development Scope



Trash Bin Status Scan Sensor (Proto)

- Displays a simple list of trash bins that require collection along with their status.



Collection Support App (Proto)

List & Status Visualization

- Bin ID & location display
- 80%+ threshold filtering
- G / Y / R status color mapping

* Sophisticated algorithms and complex payment/authorization features are omitted at the PoC level to focus on verifying information value.

Scope of Development & Use Cases

Development Scope

- 1 Implements a minimum viable function for hypothesis validation, not a fully commercialized system
- 2 Status information is categorized as Empty / Normal / Nearly Full, etc.

Use Cases

- 1 Compare outcomes with and without providing trash bin status information
- 2 Observe behavioral changes and perceived efficiency of sanitation workers and administrators

Objective

To assess whether this information is truly needed and has demand as a service, not technical completeness.